

2026 Summer Break FTL Study

Basic Lecture

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Contents

1. FEMU
2. FIO
3. Filebench
4. 논문 재현

1. FEMU

- FEMU 기반 논문 git

- ✓ <https://github.com/MoatLab/FEMU>
- ✓ <https://github.com/astlxxmu/LearnedFTL>
- ✓ <https://github.com/theXiaofu/Seg-FTL>
- ✓ devsz_mb : 용량
- ✓ net user, hostfwd=tcp::10621-:22 : 포트포워딩

run-blockbox.sh

```
#!/bin/bash
1 # Huaicheng Li <huaicheng@cs.uchicago.edu>
2 # Run FEMU as a black-box SSD (FTL managed by the device)
3
4 # image directory
5 IMGDIR=/home/choi_gunhee/LIFTL/images
6 # Virtual machine disk image
7 OSIMGF=$IMGDIR/ubuntu22.qcow2
8 #OSIMGF=$IMGDIR/LIAFS.qcow2
9
10 if [[ ! -e "$OSIMGF" ]]; then
11     echo ""
12     echo "VM disk image couldn't be found ..."
13     echo "Please prepare a usable VM image and place it as $OSIMGF"
14     echo "Once VM disk image is ready, please rerun this script again"
15     echo ""
16     exit
17 fi
18
19 sudo x86_64-softmmu/qemu-system-x86_64 \
20     -name "FEMU-BBSSD-VM" \
21     -enable-kvm \
22     -cpu host \
23     -smp 4 \
24     -m 4G \
25     -device virtio-scsi-pci,id=scsi0 \
26     -device scsi-hd,drive=hd0 \
27     -drive file=$OSIMGF,if=none,aio=native,cache=none,format=qcow2,id=hd0 \
28     -device femu,devsz_mb=16384,femu_mode=1 \
29     -net user,hostfwd=tcp::10621-:22 \
30     -net nic,model=virtio \
31     -nographic \
32     -qmp unix:./qmp-sock,server,nowait 2>&1 | tee log
```

2. FIO

- Link : https://fio.readthedocs.io/en/latest/fio_doc.html

- **Install**

- `sudo apt-get update`
`sudo apt-get install -y fio`

- **Command**

```
# 1G 순차 쓰기 FIO 명령어 sudo fio --name=seq_write_1G \ --  
filename=/dev/nvme0n1 \ --ioengine=libaio \ --iodepth=32 \ --rw=write \ --bs=128k \ -  
-direct=1 \ --size=1G \ --numjobs=1 \ --group_reporting
```

2. FIO

- Link : https://fio.readthedocs.io/en/latest/fio_doc.html
- Install
 - `sudo apt-get update sudo apt-get install -y fio`
- **Basic FIO GC Experiment**

```
sudo fio --name=fill_half \  
  --filename=/dev/nvme0n1 \  
  --ioengine=libaio \  
  --iodepth=32 \  
  --rw=write \  
  --bs=128k \  
  --direct=1 \  
  --size=10G \  
  --numjobs=1 \  
  --group_reporting
```

```
sudo fio --name=force_gc \  
  --filename=/dev/nvme0n1 \  
  --ioengine=libaio \  
  --iodepth=64 \  
  --rw=randwrite \  
  --bs=4k \  
  --direct=1 \  
  --size=10G \  
  --time_based=1 \  
  --runtime=300 \  
  --numjobs=4 \  
  --group_reporting
```

2. FIO

- Link : https://fio.readthedocs.io/en/latest/fio_doc.html

- Install

- `sudo apt-get update sudo apt-get install -y fio`

- Basic FIO GC Experiment

`sudo fio gc_test_16g.fio`

```
[global]
filename=/dev/nvme0n1
ioengine=libaio
direct=1
group_reporting
size=10G    # 16GB SSD 중 10GB(약 62.5%) 구간만 타겟팅
```

Step 1: 디스크의 10GB 구간을 순차 데이터로 채움 (Pre-conditioning)

```
[precondition]
rw=write
bs=128k
iodepth=32
numjobs=1
```

Step 2: 동일한 10GB 구간에 4KB 랜덤 쓰기를 수행하여 GC 강제 유발

```
[trigger_gc]
stonewall    # 앞의 [precondition] 작업이 완료된 후 실행
rw=randwrite
bs=4k
iodepth=64
numjobs=4    # 동시다발적인 IO 요청으로 FTL 큐 압박
time_based=1
runtime=300  # 300초 동안 10GB LBA 구간 내에서 지속적인 덮어쓰기 발생
```

3. Filebench

- Link : <https://github.com/filebench/filebench>
- Link : <https://github.com/filebench/filebench/releases>
- Install

Building Filebench from a release tarball

Download the latest release tarball from the release [page](#), unpack it, and then run traditional `configure`, `make`, and `sudo make install` commands.

- (중요) `sudo sysctl -w kernel.randomize_va_space=0`
- FS format
 - `sudo mkfs.ext4 /dev/nvme0n1`
 - `sudo mkdir -p /mnt/femu_nvme`
 - `sudo mount -o noatime /dev/nvme0n1 /mnt/femu_nvme`
 - `sudo filebench -f ./my_fileserver.f`

```
set $dir=/tmp
set $nfiles=1000
set $meandirwidth=1000000
set $filesize=cvar(type=cvar-gamma,parameters=mean:16384;gamma:1.5)
set $nthreads=16
set $iosize=1m
set $meanappendsize=16k
```

```
define fileset
name=bigfilesset,path=$dir,size=$filesize,entries=$nfiles,dirwidth=$meandirwidth,prealloc=80
```

```
define process name=filereader,instances=1
{
  thread name=filereaderthread,memsize=10m,instances=$nthreads
  {
    flowop deletefile name=deletefile1,filesetname=bigfilesset
    flowop createfile name=createfile2,filesetname=bigfilesset,fd=1
    flowop appendfilerand name=appendfilerand2,iosize=$meanappendsize,fd=1
    flowop fsync name=fsyncfile2,fd=1
    flowop closefile name=closefile2,fd=1
    flowop openfile name=openfile3,filesetname=bigfilesset,fd=1
    flowop readwholefile name=readfile3,fd=1,iosize=$iosize
    flowop appendfilerand name=appendfilerand3,iosize=$meanappendsize,fd=1
    flowop fsync name=fsyncfile3,fd=1
    flowop closefile name=closefile3,fd=1
    flowop openfile name=openfile4,filesetname=bigfilesset,fd=1
    flowop readwholefile name=readfile4,fd=1,iosize=$iosize
    flowop closefile name=closefile4,fd=1
  }
}
```

```
echo "Varmail Version 3.0 personality successfully loaded"
```

```
run 60
```

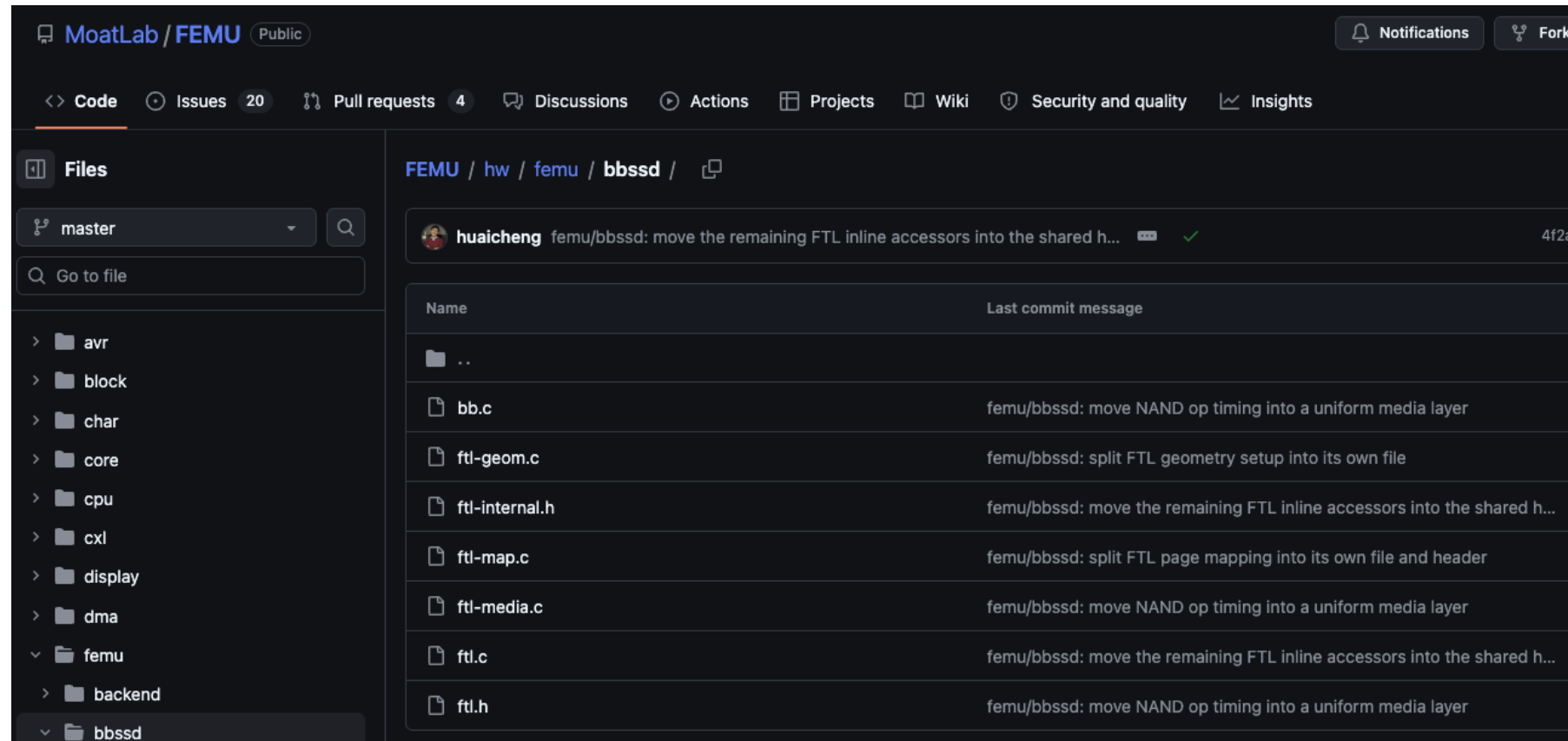
4. 할 일

- 1) 원하는 논문 한편의 실험 결과 일부를 재현해봅시다.
- 2) FEMU 소스코드 분석
- 3) 원하는 실험 진행
- 4) 논문 발표

다음 주부터

1팀은 1~3번 진행 발표
2팀 논문 발표

번갈아가면서 진행



2026 여름 방학 FTL Study

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Thank you!
Q & A ?

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